

Forward passes by hand

Project: Image-pattern Network Investigation

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Composing layers

Without opening the old lesson, explain one key idea from Composing layers. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Bias terms and decision boundaries

Without opening the old lesson, explain one key idea from Bias terms and decision boundaries. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Locate a bias omission for Forward passes by hand. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Forward passes by hand. Proposed explanation: “ReLU means the output cannot be negative”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Audit a different network independently

Use $z_1 = x + 1$, $z_2 = 2 - 2x$, $h = \text{ReLU}(z)$ component by component, and $y = 2h_1 + h_2 - 3$. Trace $x = 0$ and $x = 2$. A submitted $x = 2$ trace uses $h = [3, -2]$ and $y = 1$. Find its first error and correct all later values.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Complete a fresh row

Trace $x = -1$ through the lesson network. Record z , h and y .

My reasoning and evidence

6. W2 Locate a bias omission

At $x = 3$ a trace gives $z = [5, -1]$, $h = [5, 0]$ and $y = 5$. Identify the first incorrect stage and repair it.

My reasoning and evidence

7. W3 Check both boundaries

Trace $x = 0.5$ and $x = 2$. Which hidden score is zero in each case?

My reasoning and evidence

8. W4 Audit a different network independently

Use $z_1 = x + 1$, $z_2 = 2 - 2x$, $h = \text{ReLU}(z)$ component by component, and $y = 2h_1 + h_2 - 3$. Trace $x = 0$ and $x = 2$. A submitted $x = 2$ trace uses $h = [3, -2]$ and $y = 1$. Find its first error and correct all later values.

My reasoning and evidence

9. W5 Limit a passing test

Explain why testing only $x = 1$ in the lesson model may miss a hidden-order swap.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled forward passes by hand evidence sheet to Image-pattern Network Investigation. Show how you can record scores and activations in dependency order. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Forward passes by hand

Audit a forward pass by writing every value at its correct stage.

Toy scalar network $z=2x-1$, $h=\text{ReLU}(z)$, $y=-h+4$. Trace $x=0$ and $x=2$.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Forward passes by hand

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Preactivation

At $x=0$, $z=-1$; at $x=2$, $z=3$.



2. Activation

At $x=0$, $h=0$; at $x=2$, $h=3$.



3. Output

Outputs are 4 and 1 respectively.

CHANGE ONE THING

A submitted $x = 0$ trace uses $h=-1$ and $y = 5$.

Explain the changed result using the matching lesson.

Why should feedback identify h before correcting only y ? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

13. Project portfolio milestone

Project: Image-pattern Network Investigation

Current milestone: Frame the need. Name the user, the problem, allowed inputs and a safe simple starting method.

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Audit a forward pass by writing every value at its correct stage.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Outputs are 4 and 1 respectively.

1. Preactivation
2. Output
3. Activation

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Output Outputs are 4 and 1 respectively. A forward pass applies a fixed network to an input; it does not update weights. Keep the preactivation z , activated value h and final output y in separate trace columns. For zero, ReLU changes minus one to zero. Using the earlier minus one in the output layer is the first error, even if later arithmetic follows that mistaken value correctly.

Practice 2: explain the deciding evidence

Changed-case practice: A submitted $x = 0$ trace uses $h = -1$ and $y = 5$. Which conclusion is supported by the supplied facts?

1. Its first error is retaining the negative value after ReLU; correct $h = 0$ gives $y = 4$.
2. This one example guarantees correct results for every future input.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Its first error is retaining the negative value after ReLU; correct $h = 0$ gives $y = 4$. $\text{ReLU}(-1)=0$, so the output must use 0, not the earlier preactivation.

Repair a missing building block

Review Composing layers (ai-5/5). Goal: Trace how one layer supplies the next layer inputs. Short practice: What intermediate value explains the changed output? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Bias terms and decision boundaries (ai-5/3). Goal: Calculate a weighted score with a bias term. Short practice: Is B positive under the original or changed model? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
